HSLU Lucerne University of Applied Sciences and Arts

Computer Vision & AI

Introduction

20. Februar 2025

FH Zentralschweiz

Instructors

Thomas Koller

Joachim Wirth

Aljosa Smolic

Course Administration

Exercises:

- Quizzes on Illias
- Programming Exercises using jupyter notebooks on gpu hub (https://gpuhub.labservices.ch) and nbgrader (Introduction in first lecture)
- No ‘Testat’ on exercises, but exam they are part of the course and there might be exam questions about the exercises

Homework

- Larger Homework/Project towards the end of the semester (replacement for previous CVAI seminar)
- Presentation at the end of the semester
- Rewards points for the exam

Testat:

- Inscription to Homework

CVAI FS25: Do 20.02.2025 – Do 22.05.2025

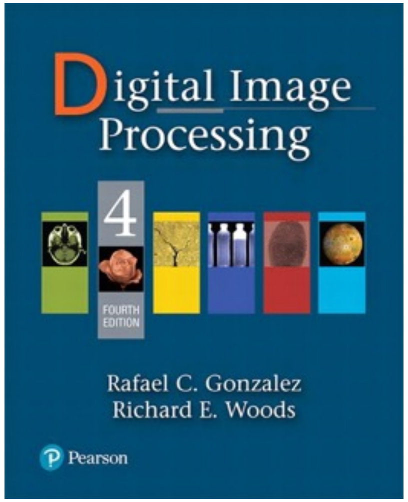
Do 12:50-15:10 und 15:30-17:50

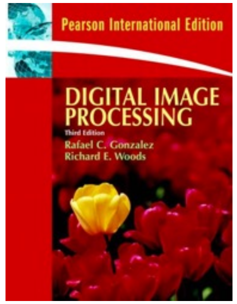
	SeWo 1 20. Feb	SeWo 2 27. Feb	SeWo 3 06. Mar	SeWo 4 13. Mar	SeWo 5 20. Mar	SeWo 6 27. Mar	SeWo 7 03. Apr
Do 1	**Thomas Koller** Einführung Digital Photography	**Joachim Wirth** Kontrasterhöhung, Glättung, Entfernen von Rauschen	**Joachim Wirth** Kompression	**Joachim Wirth** Kanten-/Linien-detektion	**Thomas Koller** Classical Image Classification, Features, SVM, Supervised Learning	**Thomas Koller** Transformer Networks, Attention, Visual Transformer	**Thomas Koller** Segmentation I: Unsupervised Learning, Clustering, Region Growing, Graph Cuts
Do 2	**Joachim Wirth** Grauwert Operationen, Morphologie	**Joachim Wirth** Farbe, Farbsysteme	**Joachim Wirth** Lineare Filter, Faltung	**Joachim Wirth** Hough Transform, Deskriptoren, SIFT,	**Thomas Koller** Image Classification with Neural Networks, Deep Learning, Conv Nets	**Thomas Koller** Object Detection, R-CNN, YOLO	**Thomas Koller** Semantic Segmentation: Fully Convolutional Neural Networks
	SeWo 8 10. Apr	**SeWo 9 17. Apr**	**SeWo 10 24. Apr**	**SeWo 11 01. Mai**	**SeWo 12 08. Mai**	**SeWo 13 15. Mai**	**SeWo 14 22. Mai**
Do 1	**Thomas Koller** Image Generation, Autoencoders GAN	Homework	I**Aljosa Smolic** Tracking: Tracking by detection, Kalman Filter, Particle Filtering	**Aljosa Smolic** 3D Reconstruction II	**Aljosa Smolic** 3D Reconstruction III	**Aljosa Smolic** 3D Reconstruction IV	**Alle** Homework Presentations
Do 2 **HSLU**	**Thomas Koller** Einführung Homework	Homework	**Aljosa Smolic** 3D Reconstruction I	Homework	Homework	Homework	**Alle** Homework Presentations

Literature

Digital Image Processing, Gonzalez & Woods, Addison Wesley, 4th International edition. (2018)

ISBN: 978-0133356724





Computer Vision: Algorithms and Applications, Richard Szeliski, 8925 pages, Springer; Auflage: 2022 ISBN: 978-3-030-34372-9

http://szeliski.org/Book/